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Yufeng Gu

Summery

Computer Engineering PhD specializing in architecture and system co-design for LLM training and inference across GPUs and custom accelerators. Research spans ***Architecture***: domain-specific accelerators, processing-in-memory, and CXL-based interconnects; ***Compiler***: PIM compilation for LLM inference across dense and MoE models, including MHA, GQA, and MLA, and ***System software***: long-context LLM inference scheduling on heterogeneous systems.

Education

University of Michigan — **Ann Arbor, MI, USA**
Ph.D., Computer Science and Engineering Department — *Sep. 2020 - Apr. 2026*

Zhejiang University (ZJU) — **Hangzhou, China**
B.Eng., College of Information Science and Electronic Engineering — *Sep. 2016 - June 2020*

Professional Experience

University of Michigan — **Ann Arbor, MI, USA**
Graduate Research Assistant, Advisor: Reetuparna Das — *Sep. 2020 – Dec. 2025*
Co-design architecture and system for large-scale emerging applications, including Generative AI and genome sequencing. Publish papers at top conferences including ISCA, ASPLOS and HPCA, awarded as Communication of ACM Research Highlights (24/10,000+) and HPCA Distinguished Artifact Honorable Mention (3/29).

Tenstorrent Inc. — **Santa Clara, CA, USA**
Performance Architect Intern, Manager: Wei-han Lien — *May 2023 – Aug. 2023*
Build a profile-guided agile performance model for Tensix AI accelerator, including computation and communication between RISC-V cores, with <10% error rate. This model reduces the performance modeling time from a few hours to ∼5 minutes, enabling fast kernel development and optimization.

Intel Labs — **Hillsboro, OR, USA**
Graduate Research Intern, Manager: Nilesh Jain — *June 2022 – Aug. 2022*
Develop an offline profile and online prediction-based resource scheduler for Quality of Service (QoS)-aware job co-location for ML inference, achieving up to 1.4× request throughput improvement over single model deployment and reducing scheduling times from ∼30 minutes to ∼10 seconds compared to state-of-the-art schedulers.

Filed Patents

[**1**] Reetuparna Das, **Yufeng Gu**. Systolic Processing-In-Memory Architecture For Large Language Model Inference. U.S. Patent Application No. 63/909,055, filed 10/31/2025. Pending.

[**2**] Reetuparna Das, **Yufeng Gu**, Alireza Khadem, Sumanth Umesh. Data Reuse-Aware Scalable Processing-In-Memory Architecture For Efficient Large Language Model Inference U.S. Patent Application No. 19/243881, filed 08/28/2025. Pending.

Under-Review Publications

[**1**] **Yufeng Gu**, Reetuparna Das. ADAPT: Adaptive Scheduling for Large Language Model Inference on Heterogeneous Server Farm *(Under Submission to ASPLOS'27)*

[**2**] **Yufeng Gu**, Samel Gobriel, Nilesh Jain, Reetuparna Das. BRIDGE: A Systolic Processing-In-Memory Architecture for Narrow GEMMs in LLM Inference *(Under Submission to IEEE Transactions on Computers)*

[**3**] **Yufeng Gu**, Vui Seng Chua, Nilesh Jain, Ravishankar Iyer, Reetuparna Das. Farm: Fast Resource Management for Quality of Service-Aware Co-Location of Machine Learning Inference. *(Under Submission to SC'2026)*

[**4**] Sumanth Umesh, Ning Liang, **Yufeng Gu**, Reetuparna Das. Addressing In-Memory Database Bottlenecks with CXL Memory Expansion. *(Under Submission to VLDB)*

[**5**] Jiazhen Wang, Finn Moore, **Yufeng Gu**, Shashwat Jaguri, Dong Eun Kim, Nicoletta Risi, Reetuparna Das. SpikeGate: An Event-Driven Neuromorphic System for Efficient Edge Vision *(Under Submission to MICRO'26)*

Peer-Reviewed Publications

* equal contribution

[1] Noah Kaplan, Jan-Niklas Schmelzle, **Yufeng Gu**, Christopher Batten, Reetuparna Das. PangenomicsBench: A Benchmark Suite and Characterization of Pangenomics. IISWC'25.

[2] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis. (Invited Paper) Communications of the ACM, 2025.

[3] Alireza Khadem*, Kamalavasan Kamalakkannan*, Zhenyan Zhu, Akash Poptani, **Yufeng Gu**, Jered Benjamin Dominguez-Trujillo, Nishil Talati, Daichi Fujiki, Scott Mahlke, Galen Shipman, Reetuparna Das. DX100: Programmable Data Access Accelerator for Indirection. (ISCA'25).

[4] **Yufeng Gu***, Alireza Khadem*, Sumanth Umesh, Ning Liang, Xavier Servot, Onur Mutlu, Ravishankar Iyer, Reetuparna Das. PIM Is All You Need: A CXL-Enabled GPU-Free System for Large Language Model Inference. (ASPLOS'25). **In Progress of Transferring to SK Hynix's Next Generation Datacenter**

[5] Alireza Khadem, Daichi Fujiki, Hilbert Chen, **Yufeng Gu**, Nishil Talati, Scott Mahlke, Reetuparna Das. Multi-Dimensional Vector ISA Extension for Mobile In-Cache Computing. (HPCA'25). **Distinguished Artifact Honorable Mention (3/29)**

[6] **Yufeng Gu**, Arun Subramaniyan, Tim Dunn, Alireza Khadem, Kuan-Yu Chen, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, Reetuparna Das. GenDP: A Framework of Dynamic Programming Acceleration for Genome Sequencing Analysis. (ISCA'23). **Communication of ACM Research Highlights (24/10,000+)**

[7] Arun Subramaniyan, **Yufeng Gu**, Timothy Dunn, Somnath Paul, Md Vasimuddin, Sanchit Misra, David Blaauw, Satish Narayanasamy, and Reetuparna Das. GenomicsBench: A Benchmark Suite for Genomics. (ISPASS'21).

[8] Xiaoxiao Li, **Yufeng Gu**, Nicha Dvornek, Lawrence H. Staib, Pamela Ventola, and James S. Duncan. Multi-site fMRI analysis using privacy-preserving federated learning and domain adaptation: ABIDE results. Medical Image Analysis 65: 101765. *IF = 11.1*

[9] Xiaoxiao Li, Yuan Zhou, Nicha C. Dvornek, **Yufeng Gu**, Pamela Ventola, and James S. Duncan. Efficient Shapley Explanation for Features Importance Estimation Under Uncertainty. (MICCAI'20).